

Equilibrium Morphologies of a Macromolecule under Soft Confinement

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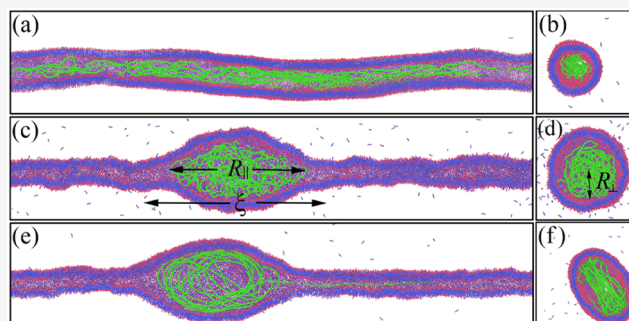


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ABSTRACT: We study equilibrium shapes and shape transformations of a confined semiflexible chain inside a soft lipid tubule by using simulations and continuum theories. The deformed tubular shapes and chain conformations depend on the relative magnitudes of their bending moduli. We characterize the collapsed macromolecular shapes by computing statistical quantities that probe the polymer properties at small length scales and report a prolate to toroidal coil transition for stiff chains. Deformed tubular shapes, calculated by using elastic theories, agree with simulations. In conjunction with scattering studies, our work provides a mechanistic understanding of macromolecular encapsulation in soft structures.



INTRODUCTION

Theoretical,^{1–5} experimental,^{6–12} and computational^{13–19} studies of polymer chains confined in a cylindrical tube have garnered a lot of current interest due to a wide range of physical applications.^{20,21} Most studies, however, focus on polymer chains under rigid confinement for which conformational statistics are well characterized by scaling theories.¹ In contrast, few studies focus on soft tubular confinement of macromolecules.^{4,22,23} Such a situation routinely arises in biophysical systems, e.g., gene transfer between bacterial pili during conjugation,^{6,10} intercellular transport of DNA and protein,^{6,24} and in bioengineering applications such as DNA confined in a lipid tether connecting two giant unilamellar vesicles (GUVs).^{8,9} In fact, nanotube-mediated transport and signaling have been implicated in various diseases, e.g., cancer and SARS COV2.^{25–29} In different situations, the membrane nanotube experiences a bulging, bead-like appearance due to the interaction between proteins, which is influenced by the mechanics of the membrane and the density of the proteins. This occurs particularly in cases where there is a varied aggregation of proteins on the membrane.^{30–35}

The statistical properties of confined macromolecules in a rigid tube are well-known.¹ From scaling theories, the axial spread of the flexible chain along the tube axis $R_{||} = ND^{-2/3}$, where N is the number of monomers and D is the diameter of the confining tube.¹ In the case of the confinement of wormlike chains (WLCs), e.g., DNA packaging in viral capsids (the persistence length of DNA is $l_p \sim 50$ nm), in cylindrical^{36–39} and spherical^{40–43} geometries has been extensively explored due to its relevance in biological and physical systems. For cylindrical confinement, the free energy of the WLC is

controlled by the persistence length l_p , the Odijk deflection length in a very strong confinement channel $l_d = D^{2/3}l_p^{1/3}$, coupled with confinement radius and tube diameter. Odijk scaling, which governs strongly confined polymers, predicts a free energy scaling as $F \sim Ll_p^{-1/3}$ when $l_p \gg R^{2/3}$. In spherical confinement, the situation becomes more complex due to increased monomer concentration, leading to stronger excluded-volume effects and competing scaling predictions for the free energy behavior $F \sim Ll_p/R^2$.⁴²

Similar scaling theories have been applied to macromolecules in soft tubular confinement, and a phase diagram characterizing confined polymer shapes as a function of the surface tension and the size of the chain has been obtained.⁴ Electrophoresis experiments on DNA confined in nanometer-sized lipid tubules⁹ reveal a “snake” to “globule” conformational transition⁴ while Monte Carlo^{22,23} and molecular dynamics⁴⁴ simulations have been used to confirm the scaling behavior. These studies, however, do not fully capture the effects of semiflexibility and the nonlinear deformations of the confining tubule on the conformational properties of the macromolecule, which sets the stage for the current work.

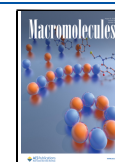
In this paper, we investigate the shape of a deformed lipid tubule encapsulating a semiflexible polymer using molecular

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dynamics simulations and continuum elastic theories. Our motivation stems from tubular structures observed in biophysical problems.^{9,24,31,32} These soft structures have elastic moduli $\sim 10\text{--}20k_B T$ such that a confining macromolecule with a size bigger than the tube dimension, $R_G \gg D_0$, is able to deform it and results in a shape change. We also study the shape transition of a homopolymer from globule to toroidal by changing the polymer persistence length l_p .

TUBULAR PROPERTIES DUE TO ENCAPSULATION OF LONG CHAIN

We consider a bilayer tubular membrane of radius R_0 and length L with a constant area $A_0 = 2\pi R_0 L$, where $R_0 \ll L$ (Figure S1). The elastic energy of deformation of the tubule is described by the Helfrich free energy⁴⁵

$$\mathcal{F} = \frac{\kappa}{2} \int_{A_0} (c_1 + c_2)^2 dA + \Sigma \int_{A_0} dA - \Delta p \int_{V_0} dV \quad (1)$$

where κ and Σ correspond to the bending modulus and the surface tension of the membrane, respectively, and Δp denotes the pressure difference acting across the membrane. The free energy is expressed in terms of the two principal curvatures, c_1 , and c_2 (assuming spontaneous curvature $c_0 \approx 0$). In the absence of an imposed pressure difference, i.e., $\Delta p = 0$, the equilibrium radius of the tubule $R_0 = \sqrt{\frac{\kappa}{2\Sigma}}$ is obtained by setting $\frac{\partial \mathcal{F}}{\partial V} = 0$.^{46,47} The deformed tubular shapes are compared to mesoscopic models of bilayer tubules with a confined semiflexible chain.

We perform coarse-grained molecular dynamics (CGMD) simulations of bilayer tubules using the LAMMPS package (SI) following the Cooke–Kremer–Deserno model^{48,49} for a tubule of length $L = 300\sigma$, and different initial radii $R_0/\sigma \approx 9, 11, 13, 15$, where σ is the unit of length. The bending modulus is obtained by computing the average radius of the tubule $\langle R \rangle$ and the tensile force in the axial direction $\kappa = F_z R_0 / 2\pi$.⁴⁹ This agrees with the value obtained by fitting the radial fluctuation spectrum.⁵⁰ The surface tension $\Sigma = \kappa / 2R_0^2$ is obtained as a consistency condition from the expression of the tether radius. A semiflexible polymer is then placed inside the tubule, and the polymer-tubule system is simulated over $t = 5\tau_{eq}$, where $\tau_{eq} = 10^6 \tau_{LJ}$ is the equilibration time. Here, τ_{LJ} is the time scale of the simulations expressed in Lennard-Jones units such that $\tau_{LJ} = \sigma \sqrt{m/\epsilon}$, where σ and m are the bead diameters and the mass of the particles, respectively. The Lennard-Jones well depth is ϵ expressed in units of $k_B T$. The equilibration time τ_{eq} is determined by monitoring the radius of gyration $R_G(t)$ of the polymer undergoing a collapse transition from a swollen to globular state and noting the time at which the $R_G(t)$ fluctuates about a mean value (SI). The total energy of the system also fluctuates around a mean value beyond τ_{eq} .

(Figure 1a,b) shows the longitudinal and transverse sections of a polymer confined in a stiff tubule with $\kappa = 24k_B T$. In this regime, i.e., $\kappa \gtrsim 20k_B T$, the entropic pressure, which is the pressure exerted by the confined polymer due to its entropy, is insufficient to deform the tube, leading to a swollen chain conformation. The reduction in the conformational freedom of the polymer imposed by spatial confinement leads to a decrease in its entropy. This entropy loss, stemming from the restricted number of accessible microstates available to the

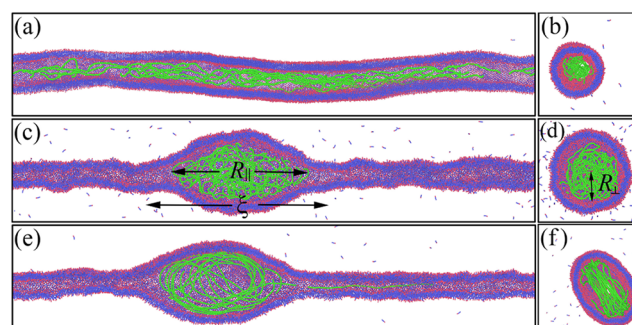


Figure 1. Equilibrium snapshots of axial and radial cross sections of a semiflexible polymer (green, $N = 2000$), confined in a bilayer lipid tubule with hydrophilic head (red) and hydrophobic tail (blue) groups obtained from CGMD simulations (SI). Swollen (a, b), and globular (c, d) chain conformations are observed for rigid $\kappa = 24k_B T$ and soft $\kappa = 12k_B T$ tubules, respectively. A prolate ellipsoidal (c, d), and a toroidal coiled (e, f) chain conformation are observed for $l_p = 13\sigma$ and $l_p \approx 200\sigma$ such that $l_p \gtrless \xi$, respectively (SI Movies).

confined polymer, manifests as an entropic cost that gives rise to the entropic pressure exerted on the tubule walls.

For softer tubes, however, where $\kappa \lesssim 12k_B T$, the polymer adopts a globular conformation. The conformation of the confined chain, as well as the shape of the bulge, depends on its persistence length l_p . For $l_p \lesssim \xi$, where ξ is the spatial deformation scale along the tube axis (Figure 1c,d), the deformed tube has a prolate shape with $R_{||} > R_{\perp}$. In contrast, for $l_p \gtrsim \xi$ (Figure 1e,f), the chain adopts a toroidal coil conformation. The bounding tube adopts a nonprolate shape (Figure 1f) with a cross-section akin to a “hockey-puck”. These conformations cannot be captured in the mean-field theories of soft macromolecular confinement.

We quantify the effect of deformation of the confined tubule on the chain size and stiffness. The time-averaged tube radius $\langle D(x)/2 \rangle$, as a function of the axial distance x , with a confined semiflexible polymer of persistence length $l_p \approx 13\sigma$ and different polymer lengths $N = 600, 800, \dots, 2600$ is measured, with data collected every 10^3 steps after equilibration. Figure 2a shows that the maximum radial bulge $\langle D_m/2 \rangle$ of the deformed tubule increases with the chain size, as $\langle D_m/2 \rangle \sim N^{0.27 \pm 0.02}$, for tubules with different initial radii R_0 and $600 \leq N \leq 7000$. Further, the maximum radial deformation $\langle D_m/2 \rangle$ is less for tubes with larger initial radii R_0 and a fixed chain size N . The deformed tubular shapes resemble a “drop wetting a fibre”⁵¹ (SI), with the axial deformation length scale $\xi \gg \langle D_m/2 \rangle$. It should be noted that the radius of the undeformed segments is less than R_0 enforcing the constant area constraint during the swollen to globule transition. There is, however, an associated increase in the volume of the tubule (Figure S11). The effects of semiflexibility on the radial deformation are analyzed by computing $\langle D_m/2 \rangle$ of a flexible polymer in a tube of initial radius $R_0 \approx 9\sigma$. As the radius of gyration of a flexible chain R_G with excluded volume interactions, scales with size as $R_G \sim N^{3/5}\sigma$ and a soft tubule deforms when $R_G \gtrsim R_0$, the lower bound of the size of a flexible chain that deforms the tubule is given by $N \sim (6R_0)^{5/3}$. While stiffer chains cause a larger radial deformation, we find that it does not alter the scaling exponent. The entropic pressure exerted by the semiflexible polymer $|\bar{p}| \equiv \Delta p / \kappa$ scales linearly with the chain length N in the strong confinement regime independent of the other parameters, e.g., R_0 , l_p , etc., as shown in the inset of Figure 2a. At equilibrium,

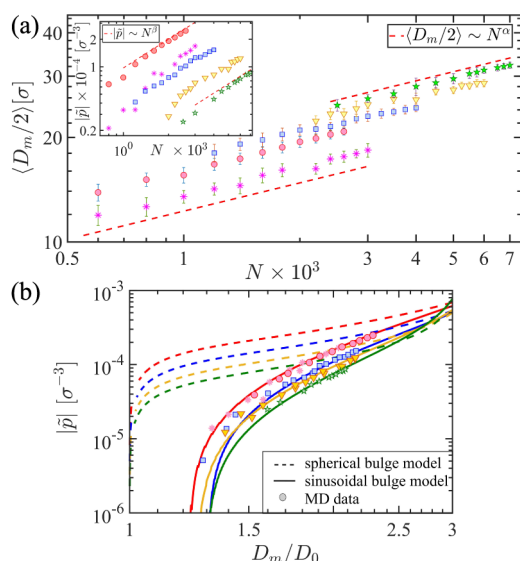


Figure 2. (a) Maximum average radial bulge $\langle D_m/2 \rangle$ vs confined chain length N for a semiflexible chain of persistence length $l_p \approx 13\sigma$ and different undeformed tubule diameter $D_0 = 18\sigma$ to 30σ . A flexible polymer in a confining tube of diameter $D_0 \approx 9\sigma$ (*) is shown for comparison. Inset shows the variation of the pressure $|p|$ exerted by the polymer on the tubule as a function of N for different radii of R_0 . (b) The pressure $|p|$ as a function of maximum bulge height of deformed tubule for two models: (i) spherical bulge and (ii) drop-on-a-fiber model. The equilibrium shape of the deformed membrane tube from MD simulation for different tube radii matches well with a drop-on-a-fiber model.

the pressure $\Delta p/k$ is equal to the mechanical pressure exerted by the walls of the confining tubule on the polymer.

Tube Deformation Models. We consider two distinct deformation shapes to analytically describe the energy of a deformed tube containing a confined polymer with a specific geometry of the deformation tubule and match the deformation tube geometry from the simulation results. The local bulge of the tubule has the geometric shape of a) a spherical bulge and b) a sinusoidal or “drop-on-a-fiber” bulge. The spherical bulge model parametrizes the membrane deformation spherical shape bulge superposed with an undeformed cylinder. In contrast, the sinusoidal bulge model represents a continuous deformation contour, avoiding a kink in the membrane in the region where the deformed and undeformed sections of the tether join. While both models capture the essential physics of the system, the “drop-on-a-fiber” parametrization offers a more physically realistic representation by smoothly connecting the deformed and undeformed regions, thereby avoiding the high curvature and associated free energy costs at the transition region.

The spherical bulge is characterized by a scaled diameter D/R_0 of a sphere (R_0 denotes the undeformed tube radius), and it is connected to the cylindrical tube of a scaled uniform radius R/R_0 . The cap segments, where the spherical bulge transitions into the cylindrical portions, are excluded from the calculations (see Figure S3b). By equating the free energy of the tubular membrane ($F_b = \pi\kappa L/R_0$, see eq S7) with the total energy of the spherical bulge connected to the cylindrical segments ($F_{\text{tot}} = F_b + F_p$, see eqs S6 and S9), the pressure $|p|$ is obtained as a function of R and D . In our simulations, the tube area is conserved in the deformed state due to the fixed number of lipid molecules. Consequently, an area constraint is applied,

resulting in distinct solutions in the $R-D$ parameter space while equating the total free energy (see Figure S6a). By minimizing the Helfrich free energy, the pressure difference parameter $|p|$ yields specific solutions for R and D , which precisely define the shape of the spherical bulge deformation. We use the equilibrium shape from our MD simulations within the analytical free energy framework to uniquely determine $|p|$, as shown by the different markers in Figure 2b. Considering the maximum bulge height D_m in the spherical bulge analytical expression model (S1), the corresponding $|p|$ suggests that higher pressure is required to achieve the specific bulge size D_m (see in Figure 2b dashed lines).

The “drop-on-a-fiber” deformation bulge is modeled as a sinusoidal shape, closely matching the deformation shape observed in our simulations. We apply a variational formulation to obtain the pressure exerted by the tubule by parametrizing the displacement from the symmetry axis x as

$$z(x) = \frac{D}{2} + \frac{R}{2} \left[1 - \cos \frac{2\pi x}{\xi} \right] \quad (2)$$

for $0 \leq z \leq \xi$. This parametrization enables the calculation of the two principal curvatures, c_1 and c_2 , as well as the volume and surface area of the deformed tubule in terms of D/R_0 , R/R_0 , and ξ . Coarse-graining the deformed tubular profiles from simulations and applying this shape parametrization, we determine the axial deformation scale of the tubule, ξ . The elastic free energy difference ΔF between the deformed tubule and a reference cylindrical state (radius R_0 and length L , with $F = 0$) is computed by assuming axisymmetric shapes and using the values c_1 and c_2 (eq S13) in eq 1. The pressure exerted by the polymer chain is determined by equating the free energy difference ΔF to the work done by the chain during tubule deformation, i.e., $\Delta F = (\Delta p/\kappa)\Delta V$, where ΔV is the change in tubule volume (see eqs S13 and S14). For a given $\Delta p/\kappa$, we obtain a family of shapes that minimize the free energy. By imposing a constant area constraint, we find a unique solution to the shape equations that minimizes the elastic free energy. For a fixed tube, the equilibrium tube radius R/R_0 and axial bulge extent ξ remain constant, while the maximum scaled bulge diameter D_m/D_0 varies with the polymer length N . As shown in Figure 2b (solid lines), the pressure difference $|p|$ yields a unique solution as a function of D_m/D_0 , which closely matches the equilibrium shapes obtained from MD simulations.

Polymer Properties in Soft Confinement. Polymer conformations inside soft tubules can be characterized on the basis of their spatial extension parallel and perpendicular to the tube axis as a function of the chain length N . Figure 3a shows the variation of the maximum radial extension $\langle R_{\perp} \rangle$ for semiflexible polymer with $l_p \approx 13\sigma$, as a function of chain length $N = 600, \dots, 7000$. A prolate globular conformation is observed, and the radial extension scales with the chain length $\langle R_{\perp} \rangle \sim N^{0.38 \pm 0.03}$ in agreement with mean-field theories.⁴ The inset in Figure 3a shows the probability distribution of the radial bulge $P(R_{\perp})$ from which the mean extension $\langle R_{\perp} \rangle$ is calculated. The free energy of the confined chain is computed $\mathcal{F} = -k_B T \ln[P(R_{\perp})]$. Differentiating the chain free energy with respect to its volume $\partial\mathcal{F}/\partial V$ yields the pressure exerted by the chain on the walls. Similar to the confining tubule, the maximum chain extension along the axial direction $R_{\parallel}$ is independent of the chain length N as shown in Figure 3b. The axial extension $R_{\parallel}$ is indeed coupled to the equilibrium

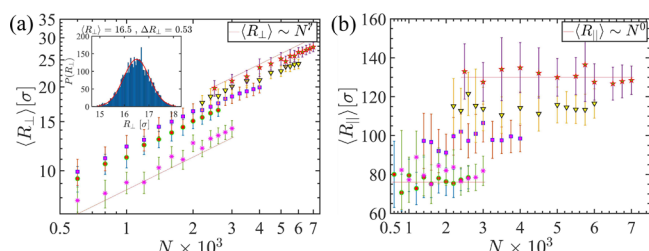


Figure 3. (a) Maximum radial extension of the polymer $\langle R_{\perp} \rangle$ vs chain size N for undeformed tube radii $R_0 \approx 9\sigma$ (•), 11σ , 13σ (▼), and 15σ (★), compared against a flexible polymer in a confining tube of radius $R_0 \approx 9\sigma$ (*). Inset shows the probability distribution $P(R_{\perp})$ of the radial bulge. (b) Maximum axial extension of polymer $\langle R_{\parallel} \rangle$ for different undeformed tube radii is independent of the chain size N .

properties of the tube radius. Specifically, the deformation bulge curvature along the tube axis cannot exceed a maximum value, as determined by our simulations. The polymer adopts an ellipsoidal crumpled conformation influenced by the bulge shape, resulting in the bulge extension along the tube length

being directly related to the axial extension of the polymer. This is in sharp contrast to the behavior of confined chains inside rigid tubes, where $R_{\parallel}$ increases linearly with chain size.

While the large scale properties of confined polymer chains can be investigated by computing $R_{\perp}$ and $R_{\parallel}$, these do not provide information about confined polymer shapes on length scales smaller than bulge deformation ξ . Figure 4 shows polymer conformations under various types of confinement. To explore the role of confining geometry, mechanical stiffness, and solvent quality on the shape of a confined polymer, we compute the (e–h) tangent–tangent correlation function $\langle \vec{t}_i \cdot \vec{t}_j \rangle$, (i–l) asphericity parameters Δ and Ξ ,⁵² and (m–p) mass distribution ρ_r as shown in Figure 4. Simulation snapshots show space-filling representations of four compact chain conformations: (a) spherical, (b) prolate, (c) toroidal coil, and (d) collapsed prolate shapes. These statistical quantities that explore the short-distance properties of the confined chain are used to characterize different compact morphologies.

The tangent–tangent correlations of a semiflexible polymer under hard spherical confinement are given by

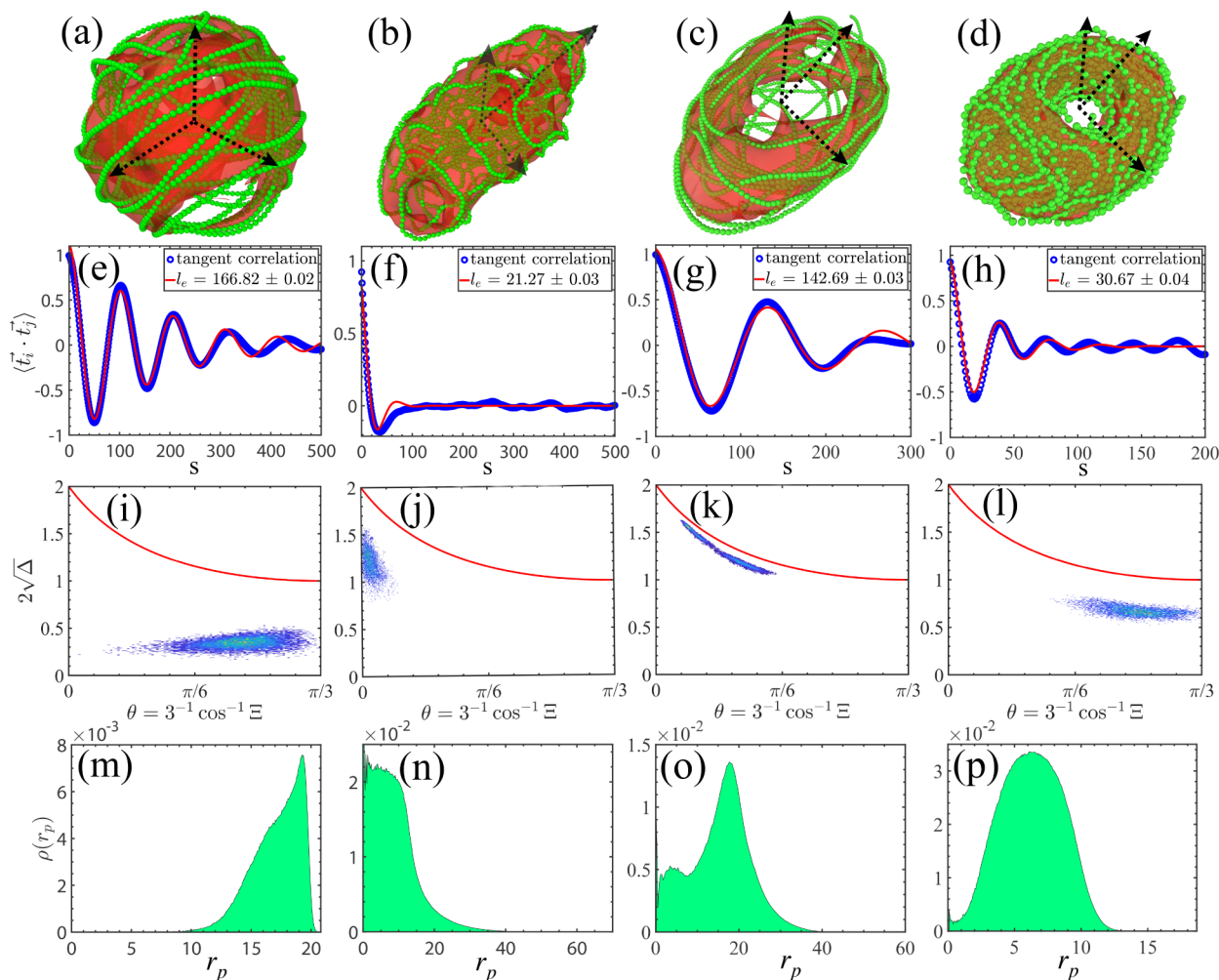


Figure 4. Conformations of semiflexible polymers: (a) inside a rigid sphere with $l_p \approx 200\sigma$, a soft tube with $\kappa \approx 12k_B T$ and (b) $l_p \approx 13\sigma$, (c) $l_p \approx 200\sigma$, and (d) $l_p \approx 200\sigma$ in a bad solvent. Panels (e)–(h) show the tangent vector correlation function of the polymer obtained from simulations (blue) and fit to a functional form (red) (see text). For stiff chains $l_p \approx 200\sigma$, a shape transition from ellipsoidal to toroidal conformation is observed, characterized by the asphericity parameters, Δ and Ξ . The distribution of monomers from the center of mass of the chain (panels (m)–(p)) is used to distinguish chain conformations based on the confining geometry, polymer stiffness, and solvent quality.

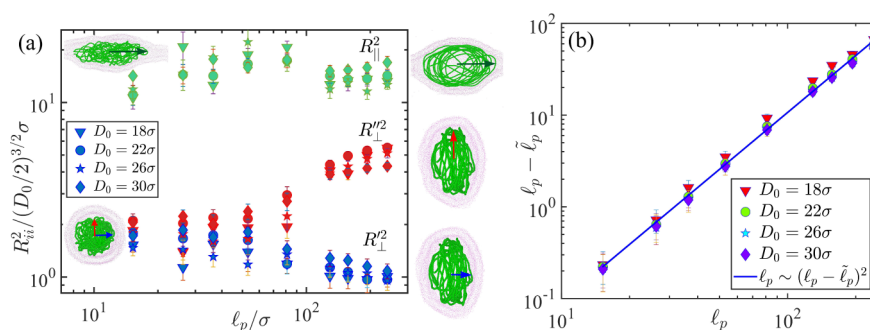


Figure 5. (a) Degenerate lifting of orthogonal gyration of polymer inside cylindrical confinement as a function of polymer persistence length l_p . A quantitative shape change from $R_{\perp}^2 \approx R_{\perp'}^2$ at lower l_p to $R_{\perp}^2 > R_{\perp'}^2$ at higher l_p . (b) Persistence length of the free polymer l_p renormalizes inside the soft confinement $\tilde{l}_p$. In the increase of l_p , the change in persistence length varies as $l_p \sim (l_p - \tilde{l}_p)^2$ from the MD simulation. The renormalized persistence lengths $\tilde{l}_p$ are calculated while polymer is inside the four different confinement radii from the MD simulation.

$$\langle \vec{t}_i \cdot \vec{t}_j \rangle \approx e^{-\frac{|i-j|}{l_e}} \left[\cos\left(\frac{s}{r_0}\right) - 2\frac{r_0}{l_e} \sin\left(\frac{s}{r_0}\right) \right] \text{ up to } O\left(\frac{r_0}{l_e}\right).^{38,42,53,54}$$

The leading order contribution to the correlation function for $\frac{l_e}{r_0} \gg 1$ is given by $\langle \vec{t}_i \cdot \vec{t}_j \rangle \approx e^{-\frac{|i-j|}{l_e}} \cos\left(\frac{s}{r_0}\right)$, which we have used to fit the tangent vector correlation data obtained from simulations. In our simulations, the radii of the confining sphere and the undeformed tube are $R_0 \approx 20\sigma$ and $R_0 \approx 9\sigma$, respectively. Figure 4e–h shows the fitted form against simulation data. The effective persistence length l_e obtained from the fit is smaller than the bare value in the free space for stiff chains. For example, Figure 4e shows the tangent-vector correlation function fitted against simulation data, which yields an effective persistence length of $l_e \sim 167\sigma$, where $r_0 \sim 19\sigma$. This can be explained simply by noting that inside the rigid spherical cavity, a stiff polymer with $l_p \sim 200\sigma \gg r_0$ wraps around the walls. As a result, the tangent vectors are decorrelated over $\pi r_0 \approx 62\sigma$, i.e., opposite points of the semicircular arc traced by the polymer. The effective persistence length l_e for semiflexible chains in contrast is greater than bare persistence length $l_p \approx 13\sigma$ under soft confinement (Figure 4f, $l_e \approx 20\sigma$, $r_0 \sim 10.5\sigma$) and in a bad solvent (Figure 4h, $l_e \approx 31\sigma$, $r_0 \sim 6.5\sigma$). Note that the increase in effective persistence length, l_e , is attributed to the strongly folded topological conformation induced by the combined effects of confinement, semiflexibility, and poor solvent conditions, which lead to a long-range correlation. The periodicity in tangent vector correlations (Figure 4e) is due to the rigid constraint enforced by the wall. A similar oscillatory behavior is observed for soft confinement of stiff polymers $l_p \gg r_0$ (Figure 4g, $l_e \approx 142\sigma$, $r_0 \sim 21\sigma$) with the polymer forming a toroidal coil. A collapsed conformation of the confined polymer in a bad solvent (Figure 4h) with $l_p \approx 13\sigma$ also shows an oscillatory behavior. This feature is absent under soft confinement (Figure 4f) where $l_p \gtrsim r_0$.

The shape transition can be quantified further using the asphericity parameters $\Delta = \frac{3}{2} \frac{\text{Tr } \hat{Q}^2}{(\text{Tr } \hat{Q})^2}$ and $\Xi = \frac{4 \text{Det } \hat{Q}}{(\frac{2}{3} \text{Tr } \hat{Q})^{3/2}}$, expressed in terms of the variance of the eigenvalues of the traceless gyration tensor about the center of mass, $\hat{Q}_{\alpha\beta} = Q_{\alpha\beta} - \frac{1}{3} \text{Tr}(Q)E$, where E is a unit tensor.⁵⁵ The

gyration tensor, $Q_{\alpha\beta} \equiv \langle (r_{\alpha} - \langle r_{\alpha} \rangle)(r_{\beta} - \langle r_{\beta} \rangle) \rangle$, where α and

$$= \frac{1}{N} \sum_i (r_i^{\alpha} - r_{\text{cm}}^{\alpha})(r_i^{\beta} - r_{\text{cm}}^{\beta})$$

β indicate components of the position vector of the i -th monomer.^{52,56} The average chain length is represented by $\bar{\lambda} = \frac{1}{3} \sum_{i=1}^3 \lambda_i$. The asphericity parameter Δ lies in the range $0 \leq \Delta \leq 1$, with $\Delta = 0$ for spheres and $\Delta = 1$ for rigid rods. The difference between the oblate and prolate shapes is captured by the parameter Ξ . As shown in Figure 4 panels (i)–(l), the heat map obtained for different configurations in the Δ – Ξ phase plane⁵⁶ is distinct for different shapes. The red line in the figures demarcates the region between accessible and inaccessible closed shapes, while the heat map provides a measure of prolate and oblate polymer conformations, which differentiate between ellipsoidal and toroidal shapes Figure 4j,k, respectively. Further, we compute the monomer distribution $\rho(|\vec{r}_p|) = 1/N \int_0^{r_p} \rho(|\vec{r}|) d\vec{r}$, as a function of r_p , where $\rho(\vec{r}) = \langle \sum_{i=0}^N \delta(\vec{r} - \vec{r}_i) \rangle$ is the local density. Figure 4m shows that $\rho(\vec{r})$ for a rigid spherical confinement of radius $R_0 \approx 20\sigma$ has a maximum at $r_p \approx R_0$ corresponding to the stiff chain wrapped inside a spherical cavity. In contrast, for soft tubular confinement (Figure 4n), the monomer distribution has a peak at $r_p \approx 0$ for globular chain conformations with stiffness $l_p \approx R_0$. For stiffer chains $l_p \gg R_0$ where the chain conformation is a toroidal coil exhibits a bimodal distribution with peaks at $r_p \approx \langle D_m/2 \rangle$ and $r_p \approx 5\sigma$ corresponding to the thickness of the toroidal shell estimated from volume conservation (see Figure 4o). A similar shift in the peak of the monomer distribution arises for collapsed chains in a bad solvent due to attractive interactions. The monomer distribution is, however, broader (Figure 4p) in comparison with the toroidal coil. These quantitative statistical measures provide a benchmark by which semiflexible polymer conformations under soft confinement can be characterized.

We previously discussed the scenario depicted in Figure 4e, where a polymer with a persistence length of $l_p \lesssim D_0$ deforms a tube, creating a sinusoidal bulge. For this bulge, the eigenvalues of the radius of gyration tensor along the tube axis, $R_{\parallel}^2$ (along the tube axis), are larger than the orthogonal eigenvalues, $R_{\perp}^2 \approx R_{\perp'}^2$ (perpendicular to the tube axis). However, as the persistence length increases beyond D_0 , the

long confined chain becomes oriented along the tube axis, resulting in a degenerate lifting of the orthogonal eigenvalues, where $R_{\perp}^2 < R_{\parallel}^2$, as depicted in Figure 5a. On the other hand, the eigenvalue along the tube axis, $R_{\parallel}^2$, remains unchanged as a function of l_p . There are several factors that can increase the persistence length of a polymer, e.g., the binding of proteins to macromolecules can increase their apparent persistence length, as demonstrated in ref 57. Adsorption of brush-like macromolecules with long side chains along with the backbone can also increase conformational deformations, as reported in ref 58. Changes in the solvent medium within the tubule can have a drastic effect on the polymer conformation, thereby altering the persistence length. To quantitatively characterize the shape transition of a morphological structure as a function of the persistence length l_p , we monitor the eigenvalues of the gyration tensor, R_{ii}^2 . When a free polymer with a persistence length $l_p \gg L$ (the confining box dimension) is confined, the effective persistence length $\tilde{l}_p$ decreases. This occurs as the angle between monomers decreases to accommodate the long, stiff chain within the narrow confinement volume. In Figure 5b, we plot the difference in persistence length between the free polymer chain and the polymer within a tubular confinement, $l_p - \tilde{l}_p$, against the persistence length of the free polymer chain. The fitting parameter from the collapsed data, displayed on a logarithmic scale, shows a behavior of $l_p \sim (l_p - \tilde{l}_p)^2$.

CONCLUSIONS

In summary, we have developed a multiscale model to characterize conformations of polymer chains under soft confinement. These confined phases are a function of the polymer chain length N , its persistence length l_p , the radius of the undeformed tube R_0 , and its bending modulus κ . For soft tubules $\kappa \lesssim 12k_B T$, we observe a swollen to globular conformational transition as the chain size N and stiffness l_p is increased. As opposed to globular structures reported in previous scaling studies,⁴ we find that a polymer adopts ellipsoidal conformations with the radial bulge of the polymer $\langle R_{\perp} \rangle$ increases with the chain size. The maximum radial bulge of the deformed bilayer tubule follows a different power law scaling with chain size N .

It is difficult to resolve macromolecular conformations inside soft tubules of nanometer dimensions using conventional scattering methods in experiments. In such situations, our calculation scheme can be used to parametrize the deformed tubular shape and obtain the principal curvatures and the change in volume (eq 1). Further, measurement of the bending modulus of the tubule κ from flicker spectroscopy allows us to compute the energy of the deformed tubule. This allows us to compute the mechanical pressure exerted by the polymer on the walls and, in turn, obtain the conformational statistics of a deformed chain in terms of the radial and axial chain deformations.

Further, we have identified statistical measures characterizing deformed polymer conformations on length scales smaller than the deformation scale in terms of the asphericity parameters, the monomer distribution, and the renormalized persistence lengths as shown in Figure 5b. These measures can

be used to distinguish deformed tubular conformations that have similar shapes on length scales comparable to axial deformation ξ . As an illustration, we report a novel ellipsoid to toroidal chain conformational transition as a function of the persistence length l_p inside soft tubes. We hope that our theoretical work will motivate experimental studies in this direction, which, in turn, will shed light on biophysical problems involving soft macromolecular confinement.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.macromol.4c01614>.

S1: Shape equation of tubules, S2: Energy estimation of different shapes of the tubule, S2.1: Spherical bulge model, S2.2: Drop on a fiber or sinusoidal bulge model, S3: Coarse-grained simulations, S4: Data analysis, S4.1: Quantifying conformational transitions, S4.2: Statistical properties of tubule, S4.3: Statistical properties of confined chains: long length scales $l \sim \xi$, S4.4: Statistical properties of confined chains: short distance $l_p \sim 1 \lesssim \xi$, and References (PDF)

SI: Movie shows the time evolution of a polymer encapsulated in a tubule starting from a random initial polymer conformation to the equilibrium state (MP4)

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Notes

The authors declare no competing financial interest.

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